Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 1 – Proof by Induction

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Prove by induction that $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$. **[5]**

2. Prove by induction that $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$. **[5]**

3. Prove by induction that $2^n > n$ for all positive integers n. **[4]**

4. Prove by induction that $n^3 - n$ is divisible by 3 for all positive integers n. **[4]**

5. Prove by induction that $\sum_{r=1}^{n} 2^{r-1} = 2^n - 1$. **[4]**

6. Prove by induction that $5^n - 1$ is divisible by 4. **[4]**

7. State the principle of mathematical induction (base case, assumption, inductive step). **[3]**

8. Prove by induction that $\sum_{r=1}^{n} r^3 = \left(\frac{n(n+1)}{2}\right)^2$. **[5]**

9. Prove that $n! > 2^n$ for $n \geq 4$. **[5]**

10. Prove by induction that $3^{2n} - 1$ is divisible by 8. **[4]**

11. Prove $\sum_{r=1}^{n}(2r-1) = n^2$. **[4]**

12. Prove that $7^n - 1$ is divisible by 6. **[4]**

13. Prove by induction that $\frac{d}{dx}(x^n) = nx^{n-1}$ for positive integers n. **[5]**

14. Prove that the sum of the interior angles of an n-gon is $(n-2)180°$. **[4]**

15. Prove $1 + 3 + 5 + \cdots + (2n-1) = n^2$ by induction. **[4]**

Section B – Hard (Q16–25)

16. Prove by induction that $\sum_{r=1}^{n} r(r+1) = \frac{n(n+1)(n+2)}{3}$. **[5]**

17. Prove that $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ where $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. **[5]**

18. Prove by induction that $n^3 + 3n^2 + 2n$ is divisible by 6. **[5]**

19. Prove that $2^{n+2} + 3^n$ is divisible by 5 for $n \geq 1$. **[5]**

20. Prove $\sum_{r=1}^{n} \frac{1}{r(r+1)} = \frac{n}{n+1}$. **[5]**

21. Prove by induction that the nth Fibonacci number satisfies $F_n \leq 2^{n-1}$. **[5]**

22. Prove that $4^n + 15n - 1$ is divisible by 9. **[5]**

23. Prove by induction that $\det(A^n) = (\det A)^n$ for a 2×2 matrix A. **[5]**

24. Prove that $n^5 - n$ is divisible by 5 for all positive integers n. **[5]**

25. Prove by induction that $\sum_{r=1}^{n} r \cdot r! = (n+1)! - 1$. **[5]**

End of Worksheet • *Show all working clearly.*